Develop Activity and Swimlane Diagram

There are many requirements modeling situations in which a **use case model may not impart information in a clear and concise manner.**

In such situations, the UML diagrams used to model the requirements are,

- **activity diagrams** and
- **swimlane diagrams.**

UML Activity diagram

An **activity diagram** is a **behavioral diagram** i.e. it depicts **the behavior** of a system.

Activity Diagrams are used to illustrate the **flow of control in a system** and refer to the steps involved in the execution of a use case.

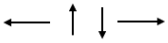
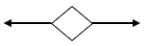
It models **sequential and concurrent activities using activity diagrams**.

An activity diagram focuses on **condition of flow and the sequence** in which it happens.

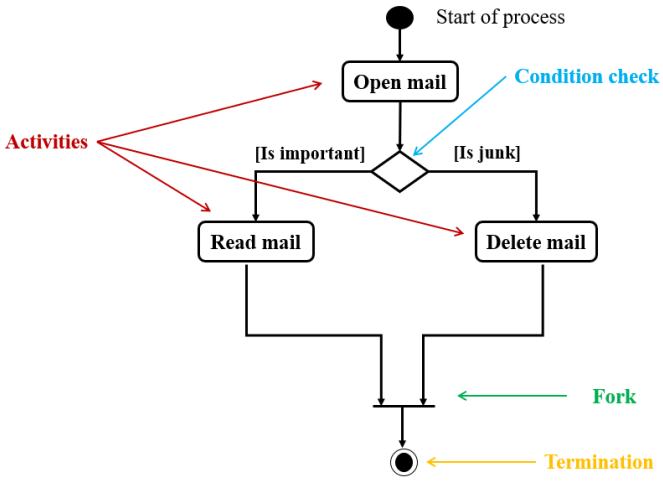
Steps to draw an activity diagram

1. Identify the initial state and the final states.
2. Identify the intermediate activities needed to reach the final state from he initial state.
3. Identify the conditions or constraints which cause the system to change control flow.
4. Draw the diagram with appropriate notations.

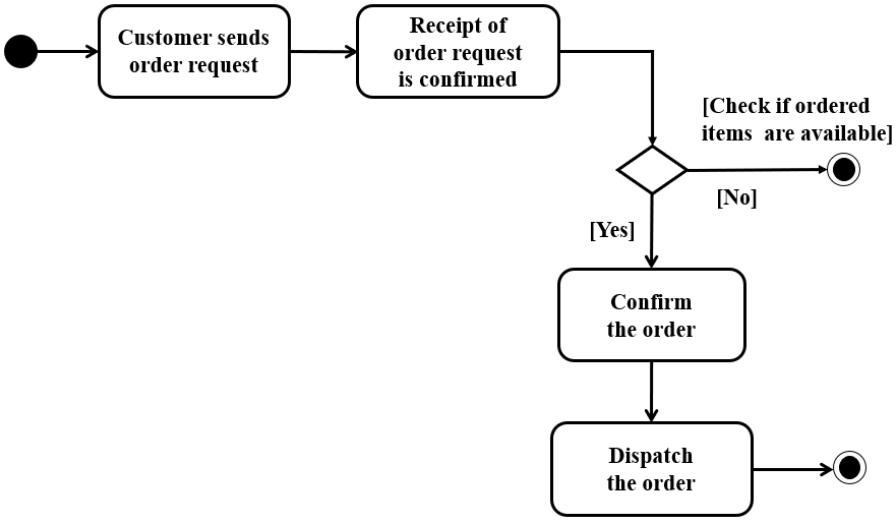
Notations in activity diagram

Notation	Meaning	Symbol
Initial State	Start of process or activity	
Rounded rectangle	specific system function/activity/process	
Arrows	action flow through the system	
Diamond	branching decisions with labels	
Fork	supports concurrent activities	
Join	concurrent activities converging into one	
Solid horizontal lines (swimlanes)	parallel activities that are occurring	
End State	Termination or End of process or activity	

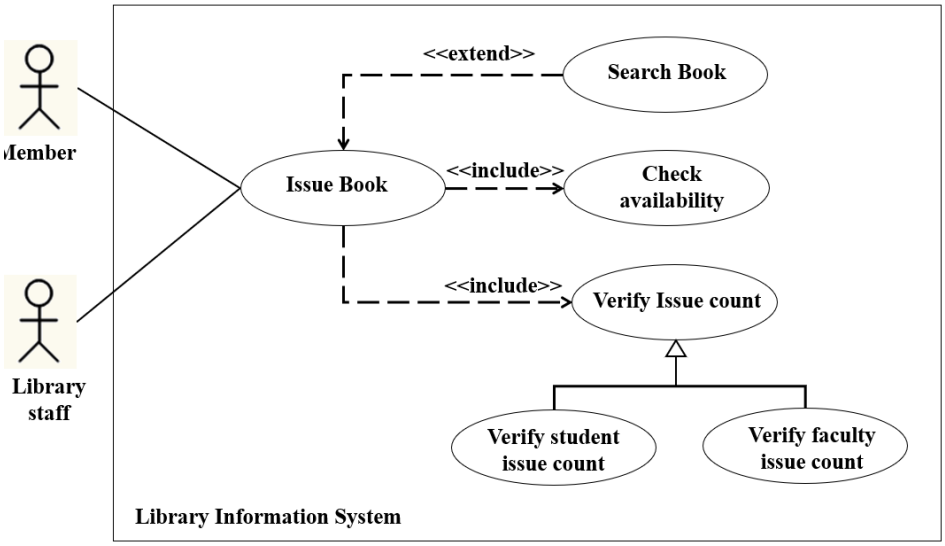
Example -activity diagram



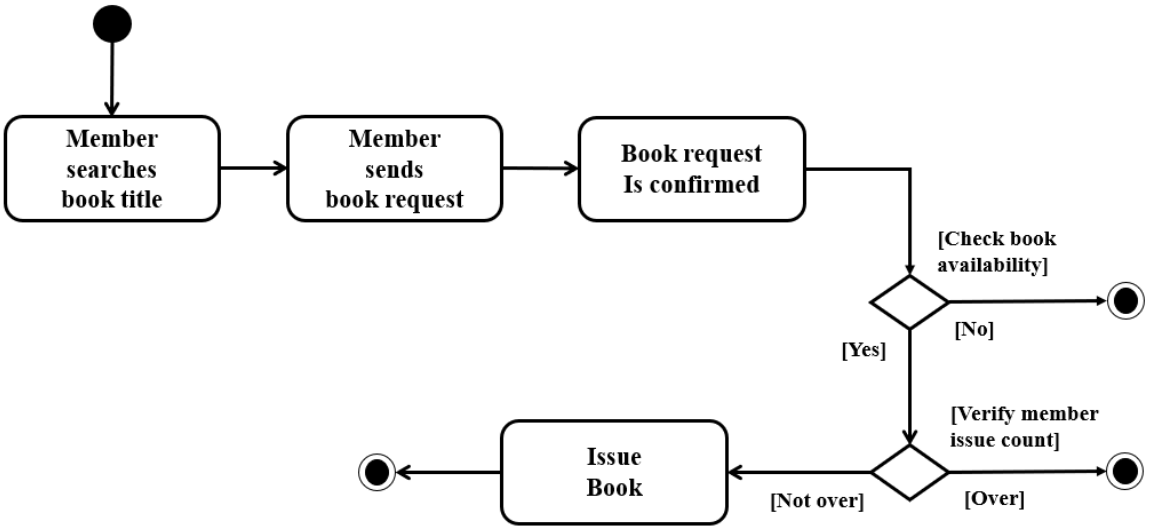
Activity diagram for order management system



Use case diagram showing "Issue book" use case



Activity diagram for "Issue book" in library information system



UML swimlane diagram

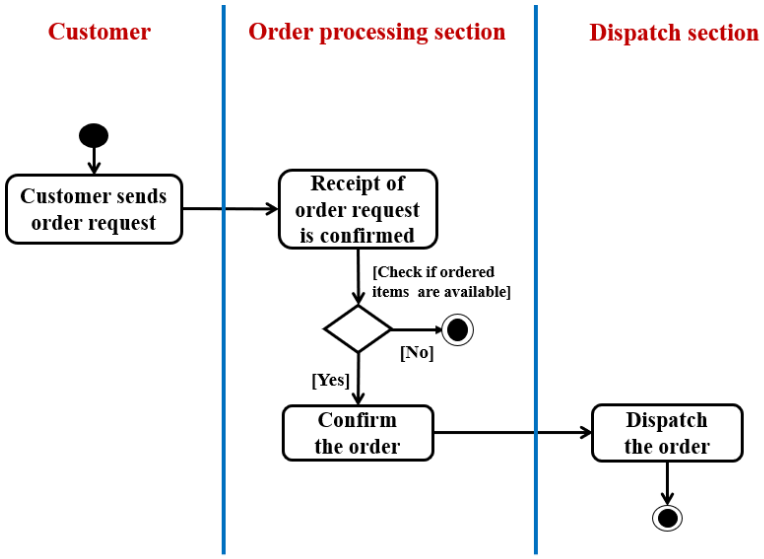
The UML swimlane diagram is a useful variation of the activity diagram and allows to represent the flow of activities described by the use case and at the same time indicate which actor (if there are multiple actors involved in a specific use case) or analysis class has responsibility for the action described by an activity rectangle.

Responsibilities are represented as parallel segments that divide the diagram vertically, like the lanes in a swimming pool.

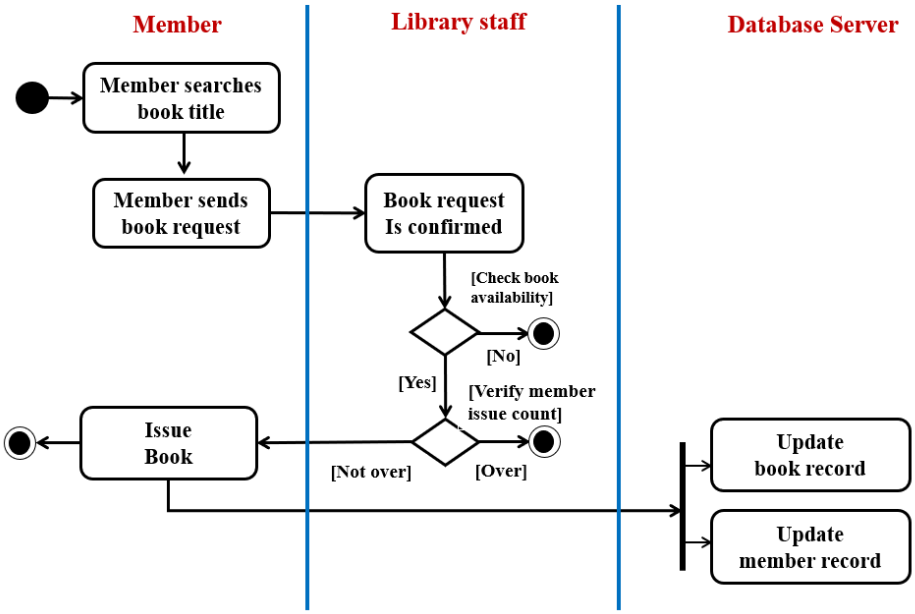
Use cases, along with the activity and swimlane diagrams, are procedurally oriented.

They represent the manner in which various actors invoke specific functions (or other procedural steps) to meet the requirements of the system.

Swimlane diagram for order management system



Swimlane diagram for library information system



https://www.tutorialspoint.com/uml/images/uml_activity_diagram.jpg

https://www.guru99.com/uml-activity-diagram.html